26/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

Regd. Office: Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - II

Answer Key

1.	(4)	37.	(1)	73.	(4)	109.	(3)	145.	(3)
2.	(2)	38.	(4)	74.	(1)	110.	(2)	146	(4)
3.	(1)	39.	(1)	75.	(1)	111.	(4)	147.	(4)
4.	(4)	40.	(2)	76.	(3)	112.	(3)	148.	(1)
5.	(3)	41.	(3)	77.	(2)	113.	(4)	149.	(3)
6.	(2)	42.	(1)	78.	(4)	114.	(1)	150.	(4)
7.	(1)	43.	(3)	79.	(4)	115.	(3)	151.	(1)
8.	(1)	44.	(2)	80.	(1)	116.	(2)	152.	(1)
9.	(3)	45.	(1)	81.	(2)	117.	(3)	153.	(3)
10.	(2)	46.	(3)	82.	(4)	118.	(1)	154.	(1)
11.	(4)	47.	(2)	83.	(2)	119.	(2)	155.	(4)
12.	(2)	48.	(2)	84.	(3)	120.	(4)	156.	(4)
13.	(3)	49.	(2)	85.	(4)	121.	(4)	157.	(2)
14.	(4)	50.	(3)	86.	(3)	122.	(3)	158.	(3)
15.	(1)	51.	(1)	87.	(4)	123.	(3)	159.	(3)
16.	(3)	52.	(3)	88.	(3)	124.	(1)	160.	(1)
17.	(1)	53.	(4)	89.	(3)	125.	(2)	161.	(2)
18.	(4)	54.	(2)	90.	(2)	126.	(4)	162.	(4)
19.	(1)	55.	(1)	91.	(2)	127.	(2)	163.	(2)
20.	(3)	56.	(2)	92.	(3)	128.	(1)	164.	(3)
21.	(4)	57.	(4)	93.	(3)	129.	(3)	165.	(3)
22.	(4)	58.	(2)	94.	(4)	130.	(2)	166.	(2)
23.	(2)	59.	(3)	95.	(2)	131.	(4)	167.	(2)
24.	(3)	60.	(3)	96.	(1)	132.	(2)	168.	(3)
25.	(2)	61.	(2)	97.	(3)	133.	(1)	169.	(4)
26.	(2)	62.	(4)	98.	(2)	134.	(3)	170.	(3)
27.	(4)	63.	(1)	99.	(3)	135.	(2)	171.	(1)
28.	(2)	64.	(2)	100.	(3)	136.	(2)	172.	(2)
29.	(4)	65.	(2)	101.	(2)	137.	(2)	173.	(1)
30.	(2)	66.	(1)	102.	(4)	138.	(1)	174.	(3)
31.	(4)	67.	(1)	103.	(3)	139.	(4)	175.	(3)
32.	(3)	68.	(3)	104.	(4)	140.	(1)	176.	(2)
33.	(4)	69.	(3)	105.	(2)	141.	(2)	177.	(1)
34.	(3)	70.	(2)	106.	(2)	142.	(3)	178.	(2)
35.	(2)	71.	(2)	107.	(1)	143.	(3)	179.	(2)
36.	(4)	72.	(4)	108.	(1)	144.	(2)	180.	(2)

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3:00 Hrs.

Test - II

Hints and Solutions

PHYSICS

1. Answer (4)

   Electron volt is the unit of energy.

2. Answer (2)

   Area under the acceleration-time graph is change in velocity.

   $v_f - v_i = \frac{1}{2} \times 20 \times 10$

   $v_f - v_i = 100$

   $v_f = 100$ m/s

3. Answer (1)

   $v = u + at$

   $v = 20 - 10 \times 3$

   $= -10$ m/s

4. Answer (4)

   $T = mg$

   It means elevator will be either at rest or will be in uniform motion.

5. Answer (3)

   The velocity of the particle will be interchanged if collision is perfectly elastic and masses are equal.

6. Answer (2)

   $mg = kx$

   $x = \frac{mg}{k}$

   In case-2

   $\frac{1}{2}kx_1^2 = 3mgx_1$

   $x_1 = \frac{6mg}{k}$

   $x_1 = 6x$

7. Answer (1)

   $X_{cm} = \frac{\int_0^L x dm}{\int_0^L dm}$

   $= \frac{\int_0^L kx^2 dx}{\int_0^L kx\,dx} \Rightarrow X_{cm} = \frac{\left[\frac{x^3}{3}\right]_0^L}{\left[\frac{x^2}{2}\right]_0^L} = \frac{2L}{3}$

8. Answer (1)

   $a = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}}$

   $a = \frac{2g\sin 30^\circ}{3}$

   $a = \frac{g}{3}$

   $v^2 = v^2 + 2as$

   $v = \sqrt{2a\ell}$

   $= \sqrt{\frac{2g\ell}{3}}$

9. Answer (3)

$W = \Delta k$

$= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 = mv^2$

$= 50 \times (10 \times 10^{-2})^2$

$W = 0.5$ J

10. Answer (2)

$y = 10\sin\left(2\pi t + \frac{\pi}{3}\right)$

$y = \frac{dy}{dt}$

$= 20\pi\cos\left(2\pi t + \frac{\pi}{3}\right)$

$= 20\pi\cos\frac{\pi}{3}$ at $t = 0$

$= 10\pi$ m/s

11. Answer (4)

$g = \frac{GM_0}{R^2}$

$R = \frac{D_0}{2}$

$g = \frac{4GM_0}{D_0^2}$

12. Answer (2)

$2\ell T = \pi R^2 \ell dg$

$R = \sqrt{\frac{2T}{\pi dg}}$

13. Answer (3)

Coefficient of thermal expansion of copper is more than iron therefore it expand more and strip bend with iron on concave side.

14. Answer (4)

$PV = nRT$

$P = nRT\left(\frac{1}{V}\right)$

15. Answer (1)

$Q = \Delta U + W$

$Q = \Delta U + \frac{Q}{3}$

$\frac{2Q}{3} = nC_v\Delta T$

$Q = \frac{3}{2}nC_v\Delta T$

$C = \frac{3}{2}\frac{R}{\left(\frac{7}{5} - 1\right)}$

$C = \frac{15R}{4}$

16. Answer (3)

$T \propto \frac{1}{\lambda}$

$\frac{T}{T_1} = \frac{2\lambda}{3\lambda}$

$T_1 = \frac{3T}{2}$

$P \propto T^4$

$P_1 \propto \left(\frac{3T}{2}\right)^4$

$\frac{P}{P_1} = \frac{16}{81}$

$P_1 = \frac{81}{16}P$

17. Answer (1)

$\frac{T_1 - T_2}{\frac{\ell}{K_1A_1}} = \frac{T_1 - T_2}{\frac{\ell}{K_2A_2}}$

$2KA_1 = KA_2$

$\frac{A_1}{A_2} = \frac{1}{2}$

18. Answer (4)

The distance between consecutive nodes is $\frac{\lambda}{2}$

$x = 2 + \frac{1}{2} = 2.5$ m

19. Answer (1)

$n = \frac{2v}{2\ell}$

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$\frac{v}{\ell} = n$

$n_1 = \frac{3v}{4\ell} = \frac{3}{4}n$

20. Answer (3)

$f = \frac{3v}{2\ell}$

$900 \times 1 = 3v$

$v = 300$ m/s

21. Answer (4)

$v = \frac{c}{\mu}$

$v = \left(\frac{3c}{5}\right)$

$t_1 = \frac{5t}{3c}$

22. Answer (4)

$M = -\frac{v}{u} = \frac{I}{O}$

$-\frac{v}{-15} = -2$

$v = -30$ cm

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{15} + \frac{1}{30} = \frac{1}{f} \Rightarrow f = -10$ cm

23. Answer (2)

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{-2} + \frac{1}{10} = \frac{1}{f}$

$\frac{1}{f} = \frac{-5+1}{10} = -2.5$ D

24. Answer (3)

When a thin metal plate is placed in the path of one of the interfering beams of light the fringes will disappear, no overlapping results.

25. Answer (2)

$I = \frac{I_0}{2}\cos^2 45 = \frac{I_0}{4}$

26. Answer (2)

$E = -\frac{dV}{dx}$

$E = -\frac{d}{dx}(4x^2 + 3x - 2)$

$E = -8x - 3$

$F = qE$

$= |2(-16 - 3)| = 38$ N

27. Answer (4)

$E = \frac{1}{2}CV^2$, $V = 6$ volt (across $4\ \mu$F)

$E = \frac{1}{2} \times 4 \times 10^{-6}(6)^2$

$E = 72\ \mu$J

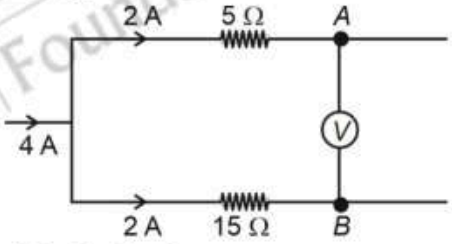
28. Answer (2)

$\frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}$

$\frac{2.4}{E_2} = \frac{120}{140}$

$E_2 = \frac{7 \times 2.4}{6} = 2.8$ V

29. Answer (4)



$V = 15 \times 2 - 5 \times 2$

$= 30 - 10 = 20$ V

30. Answer (2)

$B = \frac{\mu_0 i}{2R}\left(\frac{\phi}{2\pi}\right)$

$= \frac{\mu_0 i}{6R}$

31. Answer (4)

$M = m\ell$

$m_1\ell_1 = m_2\ell_2$

$\frac{m_1}{m_2} = \frac{\ell_2}{\ell_1}$

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32. Answer (3)

$$\Rightarrow \quad T \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$$

33. Answer (4)

$$E = -\frac{Nd\phi}{dt}$$

$$= \frac{2 \times 200 \times 500 \times 10^{-6}}{20 \times 10^{-3}} = 10 \text{ V}$$

34. Answer (3)

$$L_{eff} = \frac{L_1 L_2}{L_1 + L_2} + L_3$$

35. Answer (2)

$$(f_r) = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}, f_1 = \frac{1}{\sqrt{L/2 \times C/2}}$$

$f_1 = 2f$

36. Answer (4)

The wavelength of the matter wave does not depend on charge.

37. Answer (1)

In photoelectric phenomenon, number of photoelectrons emitted depends on intensity of light. As intensity is proportion to no. of photons emitted.

38. Answer (4)

$\lambda = \frac{h}{P}$ (hyperbola graph)

39. Answer (1)

$r \propto n^2$

$$\frac{R}{R_1} = \frac{3^2}{4^2}$$

$$R_1 = \left(\frac{16R}{9}\right)$$

40. Answer (2)

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)Z^2$$

$$\frac{1}{\lambda} = \left(\frac{1}{(1)^2} - \frac{1}{(3)^2}\right)(Z = 1)$$

$$\frac{1}{\lambda} = \frac{8R}{9}$$

41. Answer (3)

$$\frac{I_0}{3} = I_0 e^{-\lambda t}$$

$$e^{3\lambda} = 3$$

$3\lambda = \ln 3$

$$\lambda = \frac{\ln 3}{3}$$

$$I = I_0 e^{-\lambda \times 9}$$

$$I = I_0 e^{\frac{-\ln \times 3}{3} \times 9}$$

$$I = \frac{I_0}{e^{\ln 3^3}}$$

$$I = \frac{I_0}{27}$$

42. Answer (1)

$E = 931 \times 4 \approx 3724$ MeV

43. Answer (3)

At absolute zero temperature, a crystal of pure Germanium behave as a perfect insulator.

44. Answer (2)

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta} \Rightarrow \frac{1}{\alpha} - \frac{1}{\beta} = 1$$

$x - y = 1$

45. Answer (1)

$Y = \overline{A \cdot B} = A \cdot B \Rightarrow$ AND gate

CHEMISTRY

46. Answer (3)

$O_2^+(15e^-)$: Bond order = 2.5 (Paramagnetic)

$C_2^{2-}(14e^-)$: Bond order = 3.0 (Diamagnetic)

$O_2^-(17e^-)$: Bond order = 1.5 (Paramagnetic)

$O_2(16e^-)$: Bond order = 2.0 (Paramagnetic)

47. Answer (2)

% N in organic compound

$$= \frac{1.4 \times \text{meq of acid}}{\text{wt. organic compound}}$$

$$= \frac{1.4 \times 20 \times 0.5 \times 2}{0.5} = 56\%$$

48. Answer (2)

Ring of highest electron density, undergoes fastest electrophilic attack.

49. Answer (2)

Cyclic compounds which are neither aromatic nor anti-aromatic are non-aromatic in nature.

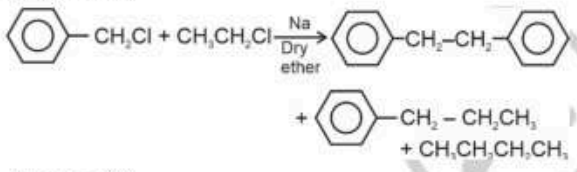
50. Answer (3)

Racemisation is observed in S_N1 mechanism, when reactant has chiral centre.

51. Answer (1)

1° alcohol reacts at slowest rate with Lucas reagent. i.e. $ZnCl_2$ + conc. HCl

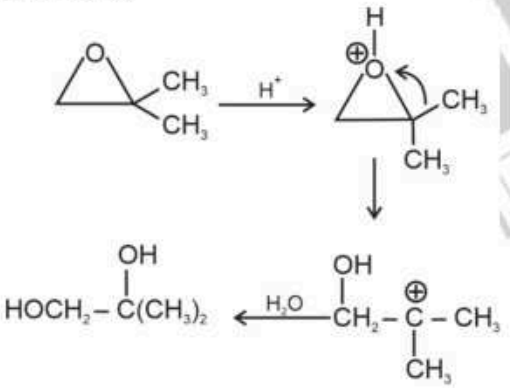
52. Answer (3)



53. Answer (4)

Radial nodes in $3d_{x^2-y^2}$ = n – l – 1 = 3 – 3 = 0

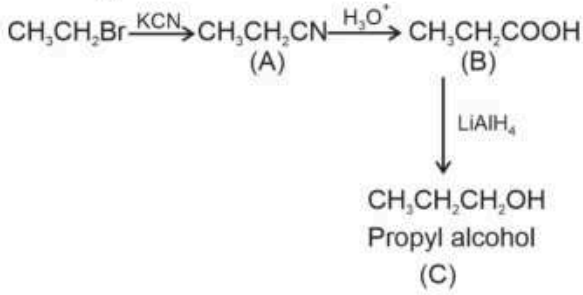
54. Answer (2)



55. Answer (1)

Glucose does not react with $NaHSO_3$.

56. Answer (2)



57. Answer (4)

Correct order of acidic nature is; NO_2CH_2COOH > HCOOH > C_6H_5COOH > CH_3COOH

58. Answer (2)

Compound having $CH_3-\overset{\overset{O}{\|}}{C}-$ and $CH_3-\overset{\overset{OH}{|}}{CH}-$ groups can give haloform test.

59. Answer (3)

$\mu = \sqrt{n(n+2)}$ BM

(n = Number of unpaired electrons)

$= \sqrt{4(4+2)}$ BM $= \sqrt{24}$ BM

60. Answer (3)

Acid stronger than H_2CO_3 will react with sodium hydrogen carbonate to give CO_2.

61. Answer (2)

In carbylamines reaction, 1° amine on reaction with chloroform in alkaline medium gives isocyanide.

62. Answer (4)

Tertiary amines do not react with benzene-sulphonyl chloride.

63. Answer (1)

Volume strength = 11.2 × molarity (H_2O_2)

5.6 = 11.2 × molarity

$$\text{Molarity} = \frac{5.6}{11.2} = 0.5$$

$$\% \text{ strength} = \frac{0.5 \times 34}{10} = 1.7$$

64. Answer (2)

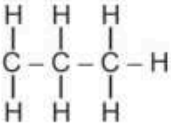
$$\text{Root mean square speed } (U_{rms}) = \sqrt{\frac{3RT}{M}}$$

$$U_{rms} \propto \sqrt{\frac{T}{M}}$$

$$\frac{(U_{rms})_2}{(U_{rms})_1} = \sqrt{\frac{\left(\frac{2T}{M/2}\right)_2}{\left(\frac{T}{M}\right)_1}} = 2$$

65. Answer (2)

'C – H' bond energy $= \frac{360}{4}$ = 90 kcal mol^{-1}



$8\,\Delta H(C-H) + 2\,\Delta H(C-C)$

So, $8\Delta H(C-H) + 2\Delta H(C-C) = 880$

$2\,\Delta H(C-C) = 880 - 720 = 160$

$\Delta H(C-C) = 80$ kcal mol^{-1}

66. Answer (1)

$$W = -2.303\,nRT\log\left(\frac{V_f}{V_i}\right)$$

$$W = 2.303 \times 2 \times R \times 300 \log\left(\frac{200}{2}\right)$$

$= -2.303 \times 2 \times R \times 300 \times 2$

$= -4 \times 2.303 \times 300\,R$

$= -1200 \times 2.303\,R$

$= -2763.6\,R$

67. Answer (1)

Formaldehyde is a common component of photochemical smog.

68. Answer (3)

Mixture of weak acid and its salt with strong base forms acidic buffer.

69. Answer (3)

CH_3COONa undergoes anionic hydrolysis.

70. Answer (2)

$$SrF_2 \rightleftharpoons Sr^{2+}(aq) + 2F^-(eq)$$

$K_{sp} = [Sr^{2+}]\,[F^-]^2$

$2.5 \times 10^{-9} = [Sr^{2+}]\,(0.1)^2 \Rightarrow Sr^{2+} = 2.5 \times 10^{-7}$

71. Answer (2)

$$(H^+)_{left} = \frac{0.1\times V\times 2 - 0.1\times 1\times V}{V+V} = \frac{(0.2-0.1)V}{2V} = 0.05\,M$$

$pH = -\log(0.05) = 2 - \log 5 = 2 - 0.7 = 1.3$

72. Answer (4)

$H_2S_2O_3 \Rightarrow 2(+1) + 2x + 3(-2) = 0,\ x = +2$

$H_2S_2O_6 \Rightarrow 2(+1) + 2x + 6(-2) = 0,\ x = +5$

$H_2S_2O_8 \Rightarrow 2(+1) + 2x + 6(-2) + 2(-1) = 0,\ x = +6$

73. Answer (4)

Species of highest positive reduction potential will easily reduce and act as good oxidising agent.

74. Answer (1)

Orbital angular momentum

$$= \frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \frac{h}{2\pi}\sqrt{6} = \sqrt{6}\,\frac{h}{2\pi}$$

75. Answer (1)

In CsCl, Cl^- are at the corners of the cube

therefore $Cl^- = \frac{1}{8}\times 8 = 1$

76. Answer (3)

As per Henry's law

$p = K_H.x$

$$1.6\text{ atm} = 1.6\times 10^3\text{ atm} \times \frac{n_{CO_2}}{n_{water}}$$

$$1.6 = 1.6\times 10^3 \times \frac{n_{CO_2}}{\frac{900}{18}}$$

$$10^{-3} = \frac{n_{CO_2}}{50} \Rightarrow n_{CO_2} = 50\times 10^{-3}$$

$= 0.05$ mol

77. Answer (2)

- $\Delta T_f = iK_f\,m = 3 \times 1.86 \times 0.1 = 0.558$
- Freezing point of solution $= 273 - 0.558 = 272.442$ K

78. Answer (4)

$$\frac{y_A}{y_B} = \frac{P_A^o X_A}{P_B^o X_B} = \frac{1000\times 2}{800\times 5} = \frac{2000}{4000} = \frac{1}{2} = 1:2$$

79. Answer (4)

$$k = \frac{2.303}{t}\log\left(\frac{A_0}{A}\right)$$

$$k = \frac{2.303}{80}\log\left(\frac{100}{20}\right) = \frac{2.303}{80}\log(5)$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{80}\log 5 \Rightarrow t_{1/2} = \frac{0.693\times 80}{2.303\log 5}$$

$= 34.4$ minutes

80. Answer (1)

$$E_{cell} = 0.0591\log\left(\frac{Ag_c^+}{Ag_a^+}\right) = 0.0591\log\left(\frac{0.1}{0.01}\right)$$

$= 0.0591$ V

81. Answer (2)

$CuSO_4$	$\longrightarrow$	Cu^{2+} +	SO_4^{2-}
Cathode :		$Cu^2 + 2e^-$ $\longrightarrow$	Cu
Anode :		$2SO_4^{2-}$ $\longrightarrow$	$S_2O_8^{2-} + 2e^-$

82. Answer (4)

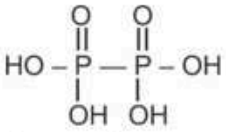
- TiO_2 sol is positive sol and for coagulation of it $[Fe(CN)_6]^{4-}$ has highest coagulating power.

83. Answer (2)

Copper matte contains Cu_2S and FeS

84. Answer (3)

Hypophosphoric Acid ($H_4P_2O_6$)



85. Answer (4)

HI > HF > HBr > HCl, (Melting point order)

86. Answer (3)

$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$

87. Answer (4)

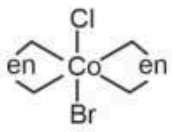
- On moving L $\rightarrow$ R, in 3*d* series, density generally increases.
- In 3*d* series Sc(Scandium) has least density.

88. Answer (3)

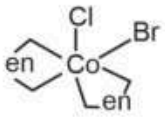
With increase in atomic number of Lanthanoids, basic character decreases.

89. Answer (3)

$[Co(en)_2(Cl)(Br)]I$

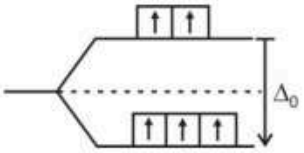


trans
(optically)
(inactive)



cis
(optically)
(active)

90. Answer (2)



CFSE = $[-0.4 \times 3 + 2 \times 0.6]\Delta_0$

= $(-1.2 + 1.2)\Delta_0 = 0$

BOTANY

91. Answer (2)

Reproduction is essential for the survival of a species.

92. Answer (3)

In five kingdom classification, plants and animals have organ level of organisation. Animals also have organ system level of organisation. Plants are autotrophic and animals are heterotrophic.

93. Answer (3)

Plants belong to family Solanaceae are dicots whereas that of poales are monocots.

94. Answer (4)

Chrysophytes do not have flagella. They float passively in water current.

95. Answer (2)

The common mode of vegetative reproduction in algae is by fragmentation.

96. Answer (1)

Nodes and internodes are found in stem. Roots lack nodes and internodes. Stems may have unicellular trichomes and may be underground. Roots also have well developed vascular system.

97. Answer (3)

In heterosporous pteridophytes, the female gametophytes retained on the parent sporophyte for variable period where the development of zygote into young embryo takes place.

98. Answer (2)

In vexillary aestivation, the two smallest anterior petals are keel.

99. Answer (3)

Ustilago belongs to class Basidiomycetes. In these fungi, karyogamy and meiosis take place in basidium.

100. Answer (3)

At maturity, sieve tube elements lack nucleus but have peripheral cytoplasm. Phloem fibres become dead at maturity.

101. Answer (2)

The cell theory proposed by Schleiden and Schwann did not explain as to how new cells were formed.

102. Answer (4)

Condensation of chromatin occurs during prophase.

103. Answer (3)

In *Amoeba* the contractile vacuole is important for osmoregulation and excretion.

104. Answer (4)

Synaptonemal complex dissolves during diplotene. The point of attachment between the homologous chromosomes i.e. chiasmata become visible after dissolution of synaptonemal complex in this phase.

105. Answer (2)

Pure water has higher water potential as compared to the solution. So, maximum amount of water will enter into the cell causing highest turgor pressure.

106. Answer (2)

Calcium is a component of middle lamella. Sulphur is main component of coenzyme A. Zinc is needed in the synthesis of auxin.

107. Answer (1)

Guttation is the water loss in liquid phase through the tip and margin of leaf blades. This process is the result of root pressure.

108. Answer (1)

Nostoc is a cyanobacteria which is also able to fix atmospheric nitrogen.

109. Answer (3)

Cornelius van Niel experimented on purple and green bacteria and demonstrated that photosynthesis is light dependent reaction in which hydrogen reduces CO_2 to carbohydrates.

110. Answer (2)

NAD^+ is reduced to NADH + H^+ during the conversion of glyceraldehyde-3-phosphate to triose bisphosphate.

111. Answer (4)

The regeneration step requires ATP molecule for phosphorylation to form RuBP.

112. Answer (3)

Ubiquinone receives electrons from both the complexes I and II and transfers to complex III.

113. Answer (4)

Trichomes are epidermal hairs on the stem.

114. Answer (1)

A sigmoid curve is a characteristic of living organism growing in a natural environment. A typical sigmoid curve is S-shaped.

115. Answer (3)

During embryogenesis, zygote undergoes mitotic cell division.

116. Answer (2)

Apical hook formation in dicot seedlings is influenced by ethylene.

117. Answer (3)

As the anther develops, the cells of the sporogenous tissue undergo meiotic division to form microspore tetrads.

118. Answer (1)

The character cannot be Y-linked as in generation II, one male individual is not showing the trait.

If the trait is autosomal recessive, the male individual in generation I will be heterozygous and if the trait is autosomal dominant, the male individual in generation I will be homozygous recessive.

119. Answer (2)

In monoecious flowering plants, male and female flowers are produced on the same plant. This prevents autogamy but not geitonogamy.

120. Answer (4)

Out of 16 individuals in F_2 generation of Mendelian dihybrid cross, two have genotype Rryy and two have genotype rrYy.

121. Answer (4)

Genes for body colour, eye colour and wing size are linked as they are present on the same chromosome *i.e.*, X-chromosome. The distance between the genes responsible for body colour and eye colour is less than that of eye colour and wing size.

122. Answer (3)

In RNA, nucleotide residue has an additional –OH group present at 2′-position in the ribose.

123. Answer (3)

Collection of germplasm and evaluation of parents are done before the process of cross hybridisation for developing new varieties.

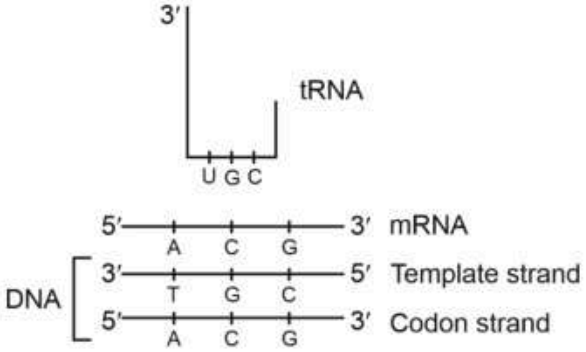
124. Answer (1)

During DNA replication, the discontinuously synthesised fragments are later joined by the enzyme DNA ligase.

125. Answer (2)

Pusa Sadabahar is a variety of chilli resistant to mosaic virus and leaf curl.

126. Answer (4)



127. Answer (2)

Saccharomyces cerevisiae is called baker's yeast as well as brewer's yeast.

128. Answer (1)

Most of the mammals and birds can tolerate and thrive in a wide range of temperatures and are called eurythermal.

129. Answer (3)

Activated sludge serve as inoculum in the aeration tank.

130. Answer (2)

Urn shaped age pyramid shows declining population in which there are fewer individuals in pre-reproductive group as compared to reproductive group.

131. Answer (4)

According to 10% law of energy transfer, sparrow has much energy as compared to other prey organisms for hawk.

132. Answer (2)

During succession, there is change in diversity of species. Therefore succession does not show stable community.

133. Answer (1)

The food chain 'Tree → Herbivorous birds → Hawk, forms spindle-shaped ecological pyramid of number.

134. Answer (3)

In scrubbing mechanism, the effluents containing sulphur dioxide are passed through a slury of water and $CaCO_3$ which combines chemically with the sulphur.

135. Answer (2)

Montreal protocol was signed to control the emission of ozone-depleting substances.

ZOOLOGY

136. Answer (2)

Ciliated epithelium are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

137. Answer (2)

Fluid filled space lacunae are present in matrix of both cartilage and bones.

138. Answer (1)

Pigments – Carotenoids, Anthocyanins

Lectins – Concanavalin A

139. Answer (4)

Maltase, lactase and sucrase are present in intestinal juice.

140. Answer (1)

Moon like face, pot belly are generally seen in childrens which are suffering from Kwashiorkar.

141. Answer (2)

Hypothalamus contains the centre for temperature regulation and for urge of eating and drinking.

142. Answer (3)

In knee-jerk reflex, the afferent neuron synapses with the efferent neuron in grey matter of spinal cord.

143. Answer (3)

The fluid filled inner ear called labyrinth consists of two parts, the bony and the membranous labyrinths. The membranous labyrinth is filled with a fluid called endolymph.

144. Answer (2)

Atlas forms pivot joint with axis. Maxilla and zygomatic bones are included in facial bones.

145. Answer (3)

Z line is present in middle of light band.

146 Answer (4)

Low level of blood Ca^{+2} causes tetany in humans. Osteoporosis may leads to high level of blood calcium.

147. Answer (4)

Glucocorticoids of adrenal cortex influence the carbohydrate metabolism in humans.

148. Answer (1)

Gastrin is a peptide hormone secreted from endocrine cells, present in gastric mucosa.

149. Answer (3)

Thymosins promotes production of anitbodies.

150. Answer (4)

Vasa recta is exclusively present in juxta-medullary nephrons.

151. Answer (1)

On an average about 650 ml of plasma is filtered by both kidneys in a minute.

152. Answer (1)

During joint diastole, the ventricles receive the blood from atria. Ventricular systole coincides with atrial diastole for about 0.3 second.

153. Answer (3)

B blood person can receives the blood from B and O blood group persons.

154. Answer (1)

Macrophages, monocytes and neutrophils are examples of phagocytic cells.

155. Answer (4)

Aorta carry oxygenated blood away from heart.

156. Answer (4)

Alary muscles are 12 pairs, present in dorsal side of body.

157. Answer (2)

All fishes are poikilotherms.

158. Answer (3)

Carcharodon belongs to class chondrichthyes.

159. Answer (3)

Neophron – Class aves

Apis – Phylum arthropoda

Chelone – Class Reptilia

Hyla, an amphibian exhibit external fertilization, *Panthera* is a viviparous mammal.

160. Answer (1)

The first cut will linearize the circular dsDNA.

The palindromic sequence for *EcoRI* is

5′ -GAATTC- 3′

3′ -CTTAAG- 5′

161. Answer (2)

Ampicillin resistance gene is present in both cloning vector pBR322 and pUC8.

162. Answer (4)

Simple stirred tank bioreactor is devoid of sparger.

163. Answer (2)

Bt crops possess property of pesticides.

164. Answer (3)

Streptokinase and tPA both are useful to dissolve the clot in disease (heart-attack).

165. Answer (3)

In 1983, Eli Lilly an American company prepared human insulin.

166. Answer (2)

Aquaculture involves production of useful aquatic plants as well as animals.

167. Answer (2)

Chicken cholera is a bacterial disease of poultry birds.

168. Answer (3)

Hisardale is a cross breed developed in Punjab.

169. Answer (4)

B-lymphocytes on activation differentiate into plasma cells which secrete antibodies.

170. Answer (3)

Thymus provide microenvironment for the maturation of T- lymphocytes.

171. Answer (1)

HIV belongs to retrovirus group.

172. Answer (2)

Dinosaurs, birds and crocodiles probably evolved from thecodonts.

173. Answer (1)

Gnetales and conifers were evolved from psilophyton.

174. Answer (3)

Australian marsupials exhibit the phenomenon of adaptive radiation.

175. Answer (3)

The number of chromosome in meiocyte of dog in 78.

176. Answer (2)

The offspring are extremely vulnerable to predators in organisms which undergo external fertilisation.

177. Answer (1)

Fluid of female genital tract is required for the capacitation of sperm.

178. Answer (2)

Developing follicles secrete estrogen hormone.

179. Answer (2)

Blastocyst undergoes gastrulation to form three germ layers, namely ectoderm, mesoderm and endoderm.

180. Answer (2)

Meiosis-II is completed in ovum only after the entry of sperm in it.

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